

AYUSH AGARWAL

ayush12345090104@gmail.com | 7888738163 | [linkedin.com/in/ayush-agarwal-09ayu](https://www.linkedin.com/in/ayush-agarwal-09ayu) | github.com/ayushagarwal120904-star

Summary

Aspiring Data Scientist with hands-on experience in Machine Learning, Deep Learning, and Generative AI. Developed AI/ML projects such as RAG chatbots and fraud detection systems using Python and TensorFlow. Strong analytical skills demonstrated through solving over 1200 data structures and algorithms problems. Seeking entry-level AI/ML or Data Science role to create scalable, data-driven solutions.

Skills

- Languages: C++, Python, C, HTML/CSS, JavaScript, SQL, R
- Technologies: Jupyter, MS Visual Studio, R Studio, Office 365 Tools, GitHub
- Interpersonal: Communication, Problem-Solving, Networking, Analytical Skills, Teamwork
- Coursework: Operating System, Computer Networks, Database Management System, Machine Learning

Projects

Retrieval-Augmented Generation (RAG) Chatbot | LangChain, Python, FAISS/ChromaDB, LLMs

July 2026

- Built a RAG chatbot that retrieves information from PDF and text documents using LangChain and an LLM.
- Implemented document loading, text chunking, and preprocessing to improve retrieval accuracy.
- Generated embeddings and stored them in FAISS/ChromaDB for efficient semantic search.
- Integrated a retrieval pipeline with conversational memory for context-aware multi-turn responses.

Telecom Fraud Detection System | TensorFlow, Python, Deep Learning, Streamlit,

Mar 2026

- Developed an end-to-end Telecom Fraud Detection system using Deep Learning (TensorFlow) achieving ~83% accuracy for binary classification of fraudulent call patterns.
- Performed data preprocessing, feature engineering, and model evaluation (train-test split) to optimize fraud prediction performance.
- Built a Streamlit-based real-time ML web application with explainable AI module to generate interpretable fraud detection insights.

Integrated Algorithmic Trading Framework | Backtrader + Binance,

Jan 2026

- Developed an end-to-end algorithmic trading framework integrating Backtrader with Binance API for seamless backtesting and live/paper trading.
- Implemented and compared Simple Linear, AR(3), and PyTorch-based Neural Network models; AR(3) achieved best performance with Sharpe Ratio 2.28.
- Built a complete trading pipeline including OHLC data processing, SMA/EMA/RSI feature engineering, automated signal generation, and realistic cost-adjusted evaluation (78.98% gross return under 3× leverage).

Toxic Comment Classifier Chat Forum | Flask, TensorFlow, Keras, LSTM, GloVe, PyMongo, MongoDB,

Mar 2025

- Developed a deep learning-based toxic comment classifier using LSTM and GloVe embeddings, integrated into a Flask-powered Q&A forum.
- Automatically detects and blocks toxic questions or replies in real-time, showing user warnings before submission.
- Designed and implemented a responsive chat interface and connected it to MongoDB using PyMongo.

Autism Spectrum Disorder (ASD) Prediction Model | Python, Scikit-learn, ML, Streamlit,

Sep 2024

- Developed an ML model to predict Autism Spectrum Disorder from behavioral and demographic data.
- Preprocessed data (normalization, encoding, outlier handling) and applied supervised algorithms (Logistic Regression, SVM, Decision Trees), evaluating with accuracy, ROC-AUC, and F1-score.

Education

Vellore Institute of Technology | Vellore, Tamil Nadu

B.Tech in Computer Science, Specialization in Data Science | Sep 2022 – June 2026 | CGPA: 8.99

Stepping Stones | Chandigarh

Intermediate Class 12th | July 2020 – July 2022 | 77%

The Gurukul | Panchkula, Haryana

Class 10th | 2020 | 89%

Experience

Panacea Infosec PVT LTD | Noida, UP • IT Intern • June 2024 – July 2024

- PCI DSS Knowledge: Understanding standards and requirements.
- Compliance Support: Assist with documentation, audits, risk assessments, and scans.
- Risk Assessment: Identify and evaluate payment data security risks.
- Technical Skills: Network Security (firewalls and segmentation), Encryption (secure data in transit and storage), Compliance & Auditing (monitor activity and automate checks).

Certifications

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Machine Learning, Data Science and Generative AI with Python
- Deep Learning 2025: Neural Networks & AI
- AI Automation in n8n: Build Agents & Voice Agents
- AI Engineer Agentic Track: The Complete Agent & MCP Course
- AI Engineer Core Track: LLM Engineering, RAG, QLORA, Agents

Achievements

- Solved +1200 DSA problems on various coding platforms (Leetcode, Codechef, etc), demonstrating strong problem-solving, algorithmic thinking, and optimization skill.
- Cleared TCS iON NQT – IT with 87.78% overall score and 94.19% in Programming (C++), May 2026.